Six-Max Preflop Equilibria via Monte Carlo CFR

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Abstract

(Abstract to be written last.)

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Notation

Symbol	Meaning
$N = \{1, \dots, n\}$, c	players; the chance player
H, Z	decision histories; terminal histories
$P(h)$, $A(h)$	player to act at h; legal actions at h
$\mathcal{I}_i$, I	player i's information partition; an information set
σ, σ^t, $\bar{\sigma}$	a behavioural strategy profile; the iterate at time t; the average strategy
$\pi^\sigma(h)$, $\pi^\sigma_i(h)$, $\pi^\sigma_{-i}(h)$	reach probability; player i's part; everyone else's (chance included)
u_i	utility of player i
$v^\sigma_i(I)$, $v^\sigma_i(I, a)$	counterfactual value of I; with a forced at I
$R^T(I, a)$, R^T_+	cumulative counterfactual regret; its positive part
$\varepsilon(\sigma)$	exploitability
$k \in \{0, \dots, 168\}$, $b(x)$, n_k	hand class; the class of combo x; combos in class k (6, 4 or 12)
eq_i	equity of player i
pot, C_i, rake(z)	pot; player i's contribution; rake taken at terminal z
ρ_i, θ	realization factor of player i; the realization parameters
S, γ, κ	stack-to-pot ratio at the flop; depth fade $\gamma = S/(S + \kappa)$; its half-life
τ	softmax temperature

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Bet sizes and game parameters

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Fitted realization parameters

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Further range charts